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EXAMINER	
LAM, VINH TANG	

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2628	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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Office Action Summary	Application No. 18/110,624	Applicant(s) KIM, Jonghyun	
	Examiner VINH T LAM	Art Unit 2628	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 02/16/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4-8,10-16 and 18-20 is/are rejected.

8) ☑ Claim(s) 3,9 and 17 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 02/16/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119 (a)-(d). The certified copy has been filed in parent Application No. **KR 10-2022-0033982**, filed on **18th, Mar. 2022**.

Claim Objections

3. Claim **13** is objected to because of the following informalities: Typo error.

Claim should **end** with **a period**.

That is "load resistance[[.]]"

Appropriate correction is required.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

4. Claim(s) **1-2, 4-8, 10-11, 13-16,** and **18-20** is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by **CHOI et al. (US Patent/PGPub. No. 20190378456)**.

Regarding Claim **1, CHOI et al.** teach **a source driver** *([0073], FIG. **1-2**, i.e. **data driving circuit 121**)* **for a display panel** *([0062], FIG. **1-2**, i.e. **display panel 110**)*, **comprising:**

an output buffer *([0085], FIG. **1-2**, i.e. **output buffer**)* **configured to output a signal** *([0077], FIG. **1-2**, i.e. **data driving timing control signal DCS including a source start pulse SSP, a source sampling clock SSC, a source output enable signal SOE**)* **to a data line** *([0073], FIG. **1-2**, i.e. **data lines DL**)* **of the display panel** *(i.e. please see above citation(s))*;

an output controller *([0077], FIG. **1-2**, i.e. **the controller 140**)* **configured to control an output** *([0077], FIG. **1-2**, i.e. **source output enable signal SOE**)* **of the output buffer** *(i.e. please see above citation(s))*;

a load resistance measuring unit *([0133], FIG. **8**, i.e. **sensing circuit 800**)* **configured to measure a load resistance** *([0134], FIG. **8**, i.e. **resistance value**)* **of at least one data line** *([0134], FIG. **8**, i.e. **at least one of the plurality of lines SL**; [0070], FIG. **8**, i.e. **SL** … may include … a **column line CSL**; [0071], FIG. **8**, i.e. **CSL may be a data line DL**)* **of the display panel** *(i.e. please see above citation(s))*; **and**

a comparison unit *([0188], FIG. **12**, i.e. **controller 130**)* **configured to compare** *([0188], FIG. **12**, i.e. **comparing**)* **the load resistance measured in the load resistance** *([0188], FIG. **12**, i.e. **first time constant related value SEN1**)* **measuring unit with an initial load resistance** *([0188], FIG. **12**, i.e. **first reference time constant value REF1**)*,

wherein the output controller *(i.e. please see above citation(s))* **is further configured to control a signal** *([0188], FIG. **12**, i.e. **a driving timing for the first line SL1**)* **to be output by the output buffer based**

on a comparison result *([0188], FIG. **12**, i.e. **Depending on a result of the comparing**)*.

Regarding Claim **2, CHOI et al.** teach **the source driver of claim 1, wherein the load resistance measuring unit** *(i.e. please see above citation(s))* **is configured to measure a first load resistance** *([0219], FIG. **15**, i.e. **first time constant related value SEN1 of the first line DL1**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of a first data line** *([0208], FIG. **15**, i.e. **first data line DL1**)* **and a second load resistance** *([0219], FIG. **15**, i.e. **second time constant related value SEN2 of the second line DL2**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of a second data line** *([0208], FIG. **15**, i.e. **second data line DL2**)* **of the display panel, wherein the output buffer** *(i.e. please see above citation(s))* **comprises a first output buffer** *([0213], FIG. **15**, i.e. **buffer BUF1**)* **configured to output a first signal** *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata (of first line)**)* **to the first data line** *(i.e. please see above citation(s))*, **and a second output buffer**

*([0213], FIG. **15**, i.e. **buffer BUF2**)* **configured to output a second signal** *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata (of second line)**)* **to the second data line,**

wherein the comparison unit *(i.e. please see above citation(s))* **is configured to compare** *([0188], [0216], FIG. **12-13 & 15**, i.e. **comparing SEN1** and **SEN2 in FIG. 13 & 15** are similar to one **SEN1 in FIG. 12**)* **the first load resistance of the first data line and the second load resistance of the second data line with the initial load resistance** *([0196], FIG. **12-13 & 15**, i.e. considered that a **time constant** of the **first line SL1 and a time constant** of the second line **SL2 are identical before the display panel 110 is stretched**)*, **respectively, and**

wherein the output controller *(i.e. please see above citation(s))* **is configured to control:**

the first output buffer *(i.e. please see above citation(s))* **based on a result** *([0206], FIG. **12-13 & 15**, i.e. **first line SL1, a signal with a larger strength** than the second line SL2 or a signal that is more over-driven)* **of comparing the first load resistance of the first data line with the initial load resistance** *(i.e. please see above citation(s))*, **and the second output**

buffer based on a result *([0206], FIG. **12-13 & 15**, i.e. **first line SL1,** a signal with a larger strength than the **second line SL2 or a signal that is more over-driven**)* **of comparing the second load resistance of the second data line with the initial load resistance** *(i.e. please see above citation(s))*.

Regarding Claim **4, CHOI et al.** teach **the source driver of claim 1, wherein the load resistance measuring unit is configured to measure the load resistance of the at least one data line of the display panel** *(i.e. please see above citation(s))* **at a time point** *([0199], FIG. **12**, i.e. **an earlier timing**)* **when the display panel is driven** *([0199], FIG. **12**, i.e. **drive the first line**)*.

Regarding Claim **5, CHOI et al.** teach **the source driver of claim 2, wherein the output controller** *(i.e. please see above citation(s))* **is configured to:**

increase a strength of a signal *([0189], FIG. **12-13 & 15**, i.e. **increased time constant**)* **to be output by the first output buffer when the first load resistance of the first data line** *(i.e. please see*

above citation(s)) **increases higher than** *([**0190**], FIG. **12-13 & 15**, i.e. **first time constant related value SEN1 is larger than the first reference time constant value REF1**)* **the initial load resistance** *(i.e. please see above citation(s))*, **and**

increase a strength of a signal *([**0193**], FIG. **13 & 14**, i.e. "**FIG. 14** shows conceptually a signal applied to the first line and a **signal applied to the second line according to the control of the driving timing shown in FIG. 13"** which means that increase in the strength of the second signal increase in second load similar to that of the first)* **to be output by the second output buffer when the second load resistance of the second data line** *(i.e. please see above citation(s))* **increases higher than the initial load resistance** *([**0190**], FIG. **12-13 & 15**, i.e. "**first time constant related value SEN1 is larger than the first reference time constant value REF1**" which means **SEN2 is similar to SEN1**)*.

Regarding Claim **6, CHOI et al.** teach **the source driver of claim 1, wherein the display panel** *(i.e. please see above citation(s))* **comprises a rolling region** *([**0250**], FIG. **18-20**, i.e. **area in which the driving integrated circuit D-IC is not located is an area at which the display panel***

110 can be stretched) **and a non-rolling region** *([0250], FIG. 18-20, i.e. driving integrated circuit D-IC area)*, **and**

wherein the load resistance measuring unit *(i.e. please see above citation(s))* **is configured to measure a load resistance** *([0134], FIG. 8, i.e. resistance value)* **of a data line** *([0134], FIG. 8, i.e. at least one of the plurality of lines SL; [0070], FIG. 8, i.e. SL … may include … a column line CSL; [0071], FIG. 8, i.e. CSL may be a data line DL)* **disposed in the rolling region** *(i.e. please see above citation(s))*.

Regarding Claim **7, CHOI et al.** teach **a display device** *([0062], FIG. 1-2, i.e. display device)*, **comprising:**

a display panel *([0062], FIG. 1-2, i.e. display panel 110)* **configured to display an image** *([0005], FIG. 1-2, i.e. mages)*;

a storage module *([0182], FIG. 1-2, i.e. memory)* **configured to store** *([0182], FIG. 1-2, i.e. stored)* **an initial load resistance** *([0188], FIG. 12, i.e. first reference time constant value REF1; [0134], FIG. 8, i.e. resistance value)* **of a data line** *([0134], FIG. 8, i.e. at least one of the plurality of lines SL; [0070], FIG. 8, i.e. SL … may include … a column line CSL; [0071], FIG. 8, i.e. CSL*

may be a data line DL *)* **of the display panel** *(i.e. please see above citation(s))*; **and**

a source driver *([0073], FIG. **1-2**, i.e. **data driving circuit 121**)* **configured to compensate** *([0179], FIG. **11-12**, i.e. **compensating**)* **a signal to be output** *([0179], FIG. **11-12**, i.e. **variance/difference in time constant**)* **according to a change** *([0179], FIG. **11-12**, i.e. **stretched**)* **in a load resistance** *([0134], FIG. **8**, i.e. **resistance value**)* **of the display panel,**

wherein the display panel *(i.e. please see above citation(s))* **comprises a plurality of data lines** *([0134], FIG. **8**, i.e. **the plurality of lines SL**; [0070], FIG. **8**, i.e. **SL** … may include … a **column line CSL**; [0071], FIG. **8**, i.e. **CSL may be a data line DL**)* **and a load resistance measuring line** *([0188], FIG. **12**, i.e. **SEN1 line**)* **for measuring at least some of the plurality of data lines** *([0188], FIG. **12**, i.e. **first line SL1**)*,

wherein the source driver *(i.e. please see above citation(s))* **comprises:**

an output buffer *([0085], FIG. **1-2**, i.e. **output buffer**)* **configured to output a signal** *([0077], FIG. **1-2**, i.e. **data driving timing control signal DCS***

including a source start pulse SSP, a source sampling clock SSC, a source output enable signal SOE) **to the plurality of data lines** *(i.e. please see above citation(s))*;

an output controller *([0077], FIG. **1-2**, i.e. **the controller 140**)* **configured to control an output** *([0077], FIG. **1-2**, i.e. **source output enable signal SOE**)* **of the output buffer** *(i.e. please see above citation(s))*;

a load resistance measuring unit *([0133], FIG. **8**, i.e. **sensing circuit 800**)* **configured to measure a load resistance** *([0134], FIG. **8**, i.e. **resistance value**)* **of the at least some of the plurality of data lines through the load resistance measuring line** *(i.e. please see above citation(s))*; **and**

a comparison unit *([0188], FIG. **12**, i.e. **controller 130**)* **configured to receive** *([0188], FIG. **12**, i.e. **comparing**)* **the initial load resistance of the at least some of the plurality of data lines from the storage module** *(i.e. please see above citation(s))* **and compare** *([0188], FIG. **12**, i.e. **comparing**)* **a load resistance** *([0188], FIG. **12**, i.e. **first time constant related value SEN1**)* **measured by the load resistance**

measuring unit with the initial load resistance *(i.e. please see above citation(s))*, **and**

wherein the output controller *(i.e. please see above citation(s))* **is further configured to control a signal** *([0188], FIG. **12**, i.e. **a driving timing for the first line SL1**)* **to be output by the output buffer to the at least some of the plurality of data lines** *(i.e. please see above citation(s))* **based on a comparison result** *([0188], FIG. **12**, i.e. **Depending on a result of the comparing**)*.

Regarding Claim **8, CHOI et al.** teach **the display device of claim 7, wherein the plurality of data lines** *(i.e. please see above citation(s))* **comprise a first data line** *([0208], FIG. **15**, i.e. **first data line DL1**)* **and a second data line** *([0208], FIG. **15**, i.e. **second data line DL2**)*, **and the load resistance measuring unit** *(i.e. please see above citation(s))* **is configured to measure a first load resistance** *([0219], FIG. **15**, i.e. **first time constant related value SEN1 of the first line DL1**; **[0134]**, FIG. **8**, i.e. **resistance value**; **[0119]-[0120]**, FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of the first data line** *(i.e. please see above citation(s))* **and a**

second load resistance *([0219], FIG. **15**, i.e. **second time constant related value SEN2 of the second line DL2**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of the second data line** *([0208], FIG. **15**, i.e. **second data line DL2**)*,

wherein the output buffer *(i.e. please see above citation(s))* **comprises a first output buffer** *([0213], FIG. **15**, i.e. **buffer BUF1**)* **configured to output a first signal** *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata**)* **to the first data line** *(i.e. please see above citation(s))*, **and a second output buffer** *([0213], FIG. **15**, i.e. **buffer BUF2**)* **configured to output a second signal** *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata**)* **to the second data line,**

wherein the comparison unit *(i.e. please see above citation(s))* **is configured to compare** *([0188], [0216], FIG. **12-13 & 15**, i.e. **comparing SEN1** and **SEN2 in FIG. 13 & 15** are similar to one **SEN1 in FIG. 12**)* **a first load resistance** *([0219], FIG. **15**, i.e. **first time constant related value SEN1 of the first line DL1**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of the first data line** *(i.e. please*

see above citation(s)) **and a second load resistance** *([0219], FIG. **15**, i.e. **second time constant related value SEN2 of the second line DL2**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of the second data line with the initial load resistance, respectively** *(i.e. please see above citation(s))*, **and**

wherein the output controller *(i.e. please see above citation(s))* **is configured to control:**

the first output buffer *(i.e. please see above citation(s))* **based on a result of comparing** *([0188], [0216], FIG. **12-13 & 15**, i.e. **comparing SEN1** and **SEN2 with REF1 in FIG. 13 & 15** are similar to one **SEN1 comparing with REF1 in FIG. 12**)* **the first load resistance of the first data line with the initial load resistance** *(i.e. please see above citation(s))*; **and**

the second output buffer *(i.e. please see above citation(s))* **based on a result of comparing** *([0188], [0216], FIG. **12-13 & 15**, i.e. **comparing SEN1** and **SEN2 with REF1 in FIG. 13 & 15** are similar to one **SEN1 comparing with REF1 in FIG. 12**)* **the second load resistance of the second data line with the initial load resistance** *(i.e. please see above citation(s))*.

Regarding Claim **10, CHOI et al.** teach **the display device of claim 7, wherein the load resistance measuring unit is configured to measure a load resistance** *([0219], FIG. **15**, i.e. **first time constant related value SEN1 of the first line DL1**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of at least one data line of the display panel at a time point** *([0017], FIG. **12**, i.e. **an earlier timing**)* **when the display panel is driven** *([0017], FIG. **12**, i.e. **drive the first line**)*.

Regarding Claim **11, CHOI et al.** teach **the display device of claim 7, wherein the storage module is configured to store information** *(i.e. please see above citation(s))* **such as numbers of the plurality of data lines** *([0195], FIG. **12-13**, i.e. **first line SL1 ... second line SL2**)* **to which load resistance measuring lines are connected** *([0197], FIG. **12-13**, i.e. **SEN1 ... and ... SEN2**)* **and numbers of the load resistance measuring lines** *([0197], FIG. **12-13**, i.e. **two lines for this example**)*.

Regarding Claim **13, CHOI et al.** teach **the display device of claim 8, wherein the output controller is configured to:**

increase a strength of a signal *([0189], FIG. **12-13 & 15**, i.e. **increased time constant**)* **to be output by the first output buffer when the first load resistance of the first data line** *(i.e. please see above citation(s))* **increases higher than** *([0190], FIG. **12-13 & 15**, i.e. **first time constant related value SEN1 is larger than the first reference time constant value REF1**)* **the initial load resistance** *(i.e. please see above citation(s))*, **and**

increase a strength of a signal *([0193], FIG. **13 & 14**, i.e. "**FIG. 14** shows conceptually a signal applied to the first line and a **signal applied to the second line according to the control of the driving timing shown in FIG. 13**" which means that increase in the strength of the second signal increase in second load similar to that of the first)* **to be output by the second output buffer when the second load resistance of the second data line** *(i.e. please see above citation(s))* **increases higher than the initial load resistance** *([0190], FIG. **12-13 & 15**, i.e. "**first time constant related value SEN1 is larger than the first reference time constant value REF1**" which means **SEN2 is similar to SEN1**)*.

Regarding Claim **14, CHOI et al.** teach **the display device of claim 7, wherein the display panel** *(i.e. please see above citation(s))* **comprises a rolling region** *([0250], FIG. **18-20**, i.e. **area in which the driving integrated circuit D-IC is not located is an area at which the display panel 110 can be stretched**)* **and a non-rolling region** *([0250], FIG. **18-20**, i.e. **driving integrated circuit D-IC area**)*, **and**

wherein the load resistance measuring line *(i.e. please see above citation(s))* **is disposed in** *(FIG. **13 &18**, i.e. as **shown** by the **figure(s)**)* **the rolling region** *(i.e. please see above citation(s))*.

Regarding Claim **15, CHOI et al.** teach **a method of operating** *([0002], FIG. **1-2**, i.e. **method of driving**)* **a source driver** *([0073], FIG. **1-2**, i.e. **data driving circuit 121**)* **to compensate** *([0179], FIG. **11-12**, i.e. **compensating**)* **a signal** *([0179], FIG. **11-12**, i.e. **variance/difference in time constant**)* **to be output to a data line** *([0134], FIG. **8**, i.e. **the plurality of lines SL**; [0070], FIG. **8**, i.e. **SL** … may include … a **column line CSL**; [0071], FIG. **8**, i.e. **CSL may be a data line DL**)* **according to a change** *([0179], FIG. **11-12**,*

i.e. ***stretched)*** **in a load resistance** ***([0134]****, FIG.* ***8****, i.e.* ***resistance value)*** **of a display panel** ***([0062]****, FIG.* ***1-2****, i.e.* ***display panel 110)*****, the method** *(i.e. please see above citation(s))* **comprising:**

measuring a load resistance ***([0134]****, FIG.* ***8****, i.e.* ***resistance value)*** **of at least one data line** ***([0134]****, FIG.* ***8****, i.e.* ***at least one of the plurality of lines SL****; [****0070****], FIG.* ***8****, i.e.* ***SL*** *… may include … a* ***column line CSL****; [****0071****], FIG.* ***8****, i.e.* ***CSL may be a data line DL*** *)* **of the display panel** *(i.e. please see above citation(s))***;**

comparing the measured load resistance *(i.e. please see above citation(s))* **with an initial load resistance** ***([0188]****, FIG.* ***12****, i.e.* ***first reference time constant value REF1****; [****0134****], FIG.* ***8****, i.e.* ***resistance value)*****; and**

controlling ***([0077]****, FIG.* ***1-2****, i.e.* ***data driving timing control signal DCS including a source start pulse SSP, a source sampling clock SSC, a source output enable signal SOE)*** **the signal to be output to** ***([0077]****, FIG.* ***1-2****, i.e.* ***source output enable signal SOE)*** **the data line** *(i.e. please see above citation(s))* **based on a comparison result** ***([0188]****, FIG.* ***12****, i.e.* ***Depending on a result of the comparing)*****.**

Regarding Claim **16, CHOI et al.** teach **the method of claim 15, wherein the measuring of the load resistance** *(i.e. please see above citation(s))* **comprises measuring a first load resistance** *([0219], FIG. **15**, i.e. **first time constant related value SEN1 of the first line DL1**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of a first data line** *([0208], FIG. **15**, i.e. **first data line DL1**)* **and a second load resistance** *([0219], FIG. **15**, i.e. **second time constant related value SEN2 of the second line DL2**; [0134], FIG. **8**, i.e. **resistance value**; [0119]-[0120], FIG. **8**, i.e. **resistance ... capacitance ... time constant**)* **of a second data line** *([0208], FIG. **15**, i.e. **second data line DL2**)* **of the display panel, wherein the comparing of the measured load resistance** *(i.e. please see above citation(s))* **comprises comparing** *([0188], FIG. **12**, i.e. **comparing**)* **the first load resistance of the first data line and the second load resistance of the second data line** *([0188], [0216], FIG. **12-13 & 15**, i.e. **comparing SEN1** and **SEN2 in FIG. 13 & 15** are similar to one **SEN1 in FIG. 12**)* **with the initial load resistance** *([0196], FIG. **12-13 & 15**, i.e. considered that a **time constant** of the **first line SL1 and a time constant** of the second line **SL2 are identical before the display panel 110 is stretched**)*, **respectively, and**

wherein the controlling of the signal *(i.e. please see above citation(s))* **comprises:**

controlling a signal to be output *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata (of fist line))*** **to the first data line** *(i.e. please see above citation(s))* **based on a result** *([0206], FIG. **12-13 & 15**, i.e. **first line SL1, a signal with a larger strength** than the second line SL2 or a signal that is more over-driven)* **of comparing the first load resistance of the first data line with the initial load resistance** *(i.e. please see above citation(s))*; **and**

controlling a signal to be output to *([0213], FIG. **15**, i.e. **outputting the image analog signals as data voltages Vdata (of second line))*** **the second data line based on a result** *([0206], FIG. **12-13 & 15**, i.e. **first line SL1,** a signal with a larger strength than the **second line SL2 or a signal that is more over-driven)*** **of comparing the second load resistance of the second data line with the initial load resistance** *(i.e. please see above citation(s))*.

Regarding Claim **18, CHOI et al.** teach **the method of claim 15, wherein the measuring of the load resistance** *(i.e. please see above citation(s))* **is performed at a time point** *([0199], FIG. **12**, i.e. **an earlier timing**)* **when the display panel is driven** *([0199], FIG. **12**, i.e. **drive the first line**)*.

Regarding Claim **19, CHOI et al.** teach **the method of claim 15, further comprising:**

storing *([0183], FIG. **12**, i.e. **controller 130 may store the time constant related value**)* **the measured load resistance,**

wherein the controlling of the signal *(i.e. please see above citation(s))* **is performed when it is determined that a signal** *([0188], FIG. **12**, i.e. **a driving timing for the first line SL1**)* **to be output to the data line is changed** *([0179], FIG. **11-12**, i.e. **stretched**)* **in the load resistance comparison step** *(i.e. please see above citation(s))*.

Regarding Claim **20, CHOI et al.** teach **the method of claim 15, wherein the controlling of the signal is performed when the**

source driver *(i.e. please see above citation(s))* **outputs a signal** *([0077], FIG.* ***1-2****, i.e.* ***data driving timing control signal DCS including a source start pulse SSP, a source sampling clock SSC, a source output enable signal SOE)*** to a rolling region** *([0250], FIG.* ***18-20****, i.e.* ***area in which the driving integrated circuit D-IC is not located is an area at which the display panel 110 can be stretched)*** of the display panel** *(i.e. please see above citation(s))*.**

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claim(s) **12** is/are rejected under 35 U.S.C. 103 as being unpatentable over **CHOI et al. (US Patent/PGPub. No. 20190378456)** in view of **Yildiz et al. (US Patent/PGPub. No. 11024224).**

Regarding Claim **12, CHOI et al.** teach **the display device of claim 11.**

However, **CHOI et al.** do not explicitly disclose

the storage module is configured as a flash memory or an EPROM.

In the same field of endeavor, **Yildiz et al.** disclose

the storage module *(Col. **9**, Ln. **4-7**, FIG. **2**, i.e. **controller 32**)* **is configured as a flash memory** *(Col. **9**, Ln. **4-7**, FIG. **2**, i.e. **code stored in flash memory**)* **or an EPROM.**

It would have been obvious to a person having ordinary skill in the art at the time the invention's effective date was filed to recognize that **CHOI et al.'s [0082]** disclosing of micro control unit MCU that would obviously include embedded flash memory for storing executable instructions and pertinent values/parameters for calculating, comparing, and compensating stretched display as disclosed by **Yildiz et al.** of a processor embedded with flash memory.

Allowable Subject Matter

6. Claim(s) **3, 9,** and **17** is/are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

7. The following is an examiner's statement of reasons for allowance:

CHOI et al. (US Patent/PGPub. No. 20190378456) teach a display device includes a stretchable display panel that includes a plurality of lines and a plurality of

subpixels; a driving circuit that drives the plurality of lines; a controller that controls the driving circuit; and a sensing circuit that sends a time constant related value including at least one of a resistance value, a capacitance value and a time constant value of the plurality of lines, and that outputs the sensed time constant related value, wherein the controller controls a timing for driving at least one of the plurality of lines, or controls characteristics of a signal supplied to at least one of the plurality of lines to be changed in accordance with the time constant related value obtained by sensing at least one of the plurality of lines when the stretchable display panel is stretched.

Liu (US PGPUB./Pat. No. 10268253) teaches a flexible display includes a display panel, a gate driver, and a bending sensing unit. The display panel has a plurality of rows of pixels arranged in sequence. The gate driver has a plurality of stages of shift register circuitries electrically coupled to each other in sequence. The shift register circuitries are electrically coupled to the plurality of rows of pixels, and are configured to provide a plurality of refreshing pulses for driving the plurality of rows of pixels to refresh. Each stage of the shift register circuitries are further configured to provide a trigger pulse to drive another stage of the shift register circuitries. The bending sensing unit is electrically coupled to the gate driver. The bending sensing unit is configured to control, according to bending of the display panel, the impedance of a path used by at least one of the stages of the shift register circuitries to transmit the trigger pulse to the other stage of the shift register circuitries.

The subject matter of the independent claims could either not be found or was not suggested in the prior art of record. The subject matter not found was a display device including

"…control a signal to be output by the output buffer to at least one data line disposed between the first data line and the second data line as an interpolated output based on the first load resistance of the first data line and the second load resistance of the second data line…" (Claim **3**; Claims **9** and **17** are similar), in combination with the other elements (or steps) of the device or apparatus and method recited in the claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to VINH TANG LAM whose telephone number is (571) 270-3704. The examiner can normally be reached Monday to Friday 8:00 AM to 5:00 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nitin K Patel can be reached on (571) 272-7677. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/VINH T LAM/
Primary Examiner, Art Unit 2628